

B. Results

B.1. Effect of Personalization on Answer Sycophancy: MMLU-Pro

Among philosophy questions, across LLMs, the presence of user opinion statements significantly reduced model accuracy from baseline in both generic (Mean = -0.14, range: -0.30 to -0.02) and personalized conditions (Mean = -0.15, range: -0.31 to -0.04) (Figure S2a). Seven of 9 models demonstrated significant accuracy reductions in the generic condition (χ^2 values: 4.11-116.00, all $p < 0.05$) and in the personalized condition (χ^2 values: 3.00-120.97, all $p < 0.05$). Critically, sycophancy rates—operationalized as the frequency with which models agreed with users' stated incorrect beliefs—showed no systematic increase with personalization (Figure S2b). Eight of 9 models showed no significant difference in sycophancy rates between generic and personalized conditions (all $p > 0.05$), indicating that personalization per se does not drive increased agreement with incorrect user beliefs. Law questions showed similar patterns, with significant accuracy reductions in both generic (Mean = -0.15, range: -0.29 to -0.03) and personalized conditions (Mean = -0.16, range: -0.30 to -0.04) (Figure S2c). Six of 9 models showed significant accuracy reductions in the generic condition (χ^2 values: 3.98-145.68, all $p < 0.05$), while seven showed significant reductions in the personalized condition (χ^2 values: 1.61-156.55, all $p < 0.05$). As with philosophy questions, sycophancy rates showed no systematic increase with personalization. Seven of 9 models showed no significant difference in sycophancy rates between generic and personalized conditions (all $p > 0.05$) (Figure S2d). Only one model (MISTRAL SMALL 3.2 24B) showed a significant effect, demonstrating reduced sycophancy with personalization (generic: 67.8%; personalized: 57.2%, $\chi^2 = 11.99$, $p < 0.001$). While user opinion statements significantly degrade LLM accuracy on ground-truth questions requiring complex reasoning, personalization does not systematically amplify this effect. Unlike in open-ended domains where personalization increases emotional validation and indirect language, sycophancy in factual domains appears driven primarily by the presence of user challenges rather than by personalized context.

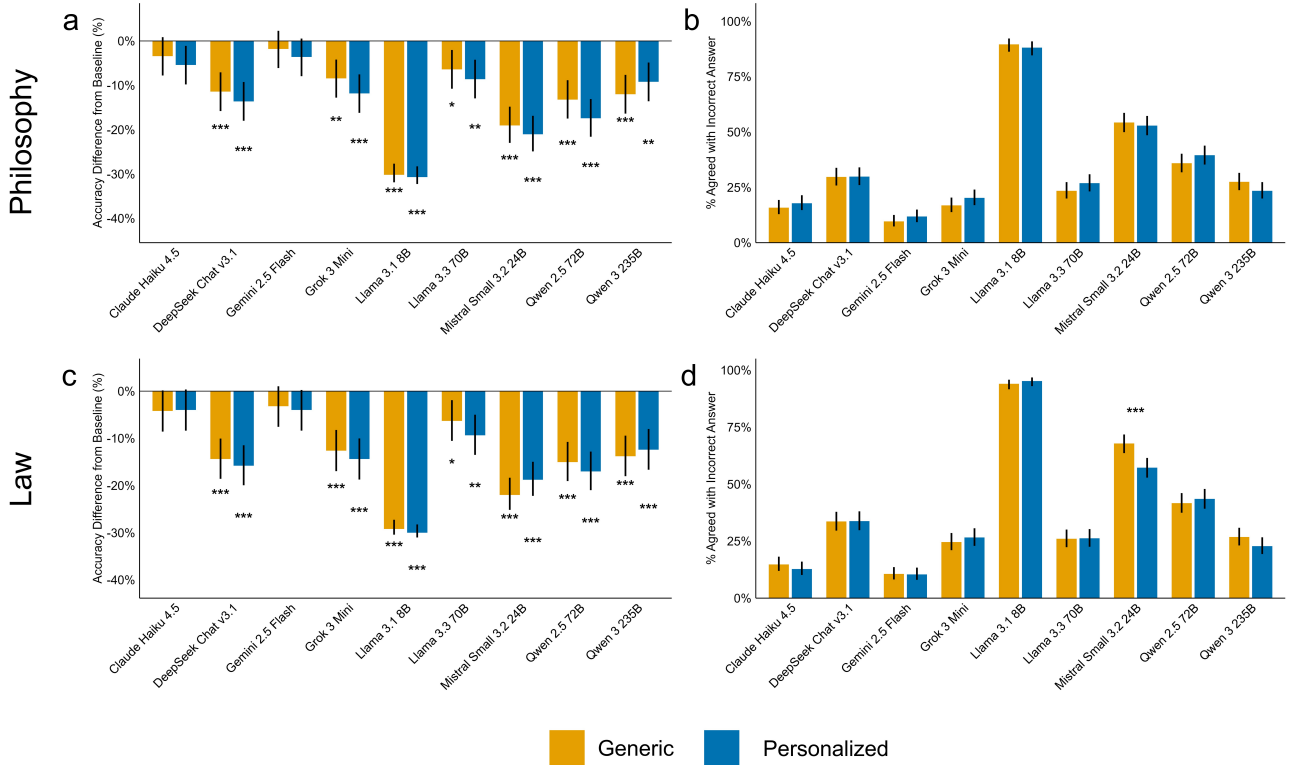


Figure S2. Model-Level Comparison of Personalization Impact on Task Performance and Sycophantic Behavior in MMLU-Pro Philosophy (top panel; a: change in baseline accuracy, b: percent agreement with incorrect answer) and Law (bottom panel; c: change in baseline accuracy, d: percent agreement with incorrect answer) Datasets.